2.9.1 Radians and angular displacement

Angular displacement describes rotation about a centre. In radians it equals arc length divided by radius, with one complete turn equal to 2π rad. Convert angles to radians when using arc-length or circular-motion equations. The same angle at different radii gives different arc lengths, so angular and linear displacement are not interchangeable.

$$\theta = \frac{s}{r}$$

θ: angular displacement (rad); s: arc length (m); r: radius (m).

Angular speed, linear speed and period

Angular speed is the angle in radians swept out per second. Linear speed is distance travelled along the circle per second. At fixed angular speed, a larger radius gives greater linear speed. A complete uniform revolution covers 2πr and 2π rad, allowing both speeds to be found from the period. Their units differ.

$$\omega = \frac{\Delta\theta}{\Delta t}$$
$$v = r\omega$$
$$T = \frac{2\pi}{\omega} = \frac{2\pi r}{v}$$

Uniform circular motion: ω is angular speed (rad s^{-1}); Δθ: swept angle (rad); Δt: time (s). Otherwise the ratio gives average angular velocity.

v: tangential speed (m s^{-1}); r: radius (m); ω: angular speed (rad s^{-1}).

Uniform circular motion: T: period (s); ω: rad s^{-1}; r: m; v: m s^{-1}.

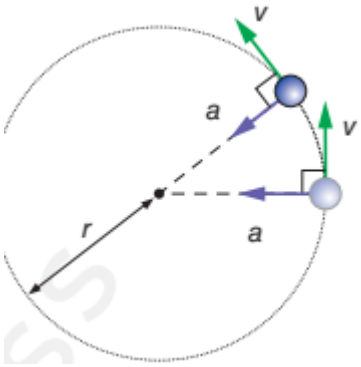
Centripetal acceleration and resultant force

In uniform circular motion, speed is constant but velocity continually changes direction. The acceleration therefore points towards the centre. The required radial resultant is called centripetal force. It is supplied by real forces such as tension, gravity or friction. Draw the real forces in the free-body diagram, then find their radial resultant.

$$a_c = \frac{v^2}{r} = r\omega^2$$
$$F_c = \frac{mv^2}{r} = mr\omega^2$$

Centripetal acceleration a_c (m s^{-2}) points towards the centre. v: speed (m s^{-1}); r: radius (m); ω: rad s^{-1}.

F_c: inward resultant force (N), not an additional force; m: kg; v: m s^{-1}; r: m; ω: rad s^{-1}.



Velocity is tangential to the circle, while centripetal acceleration points towards its centre.